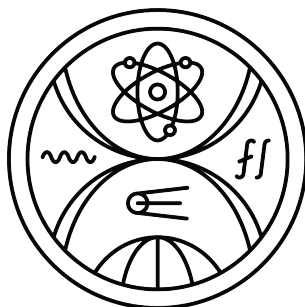


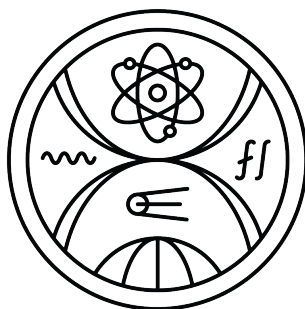
UNIVERZITA KOMENSKÉHO V BRATISLAVE
FAKULTA MATEMATIKY, FYZIKY A INFORMATIKY



AKTÍVNA PERCEPCIA V ROBOTICKOM VIDENÍ

Diplomová práca

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Študijný program: Aplikovaná informatika
Študijný odbor: Aplikovaná informatika
Školiace pracovisko: Katedra aplikovanej informatiky
Vedúci práce: prof. Ing. Igor Farkaš, Dr.



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

ZADANIE ZÁVEREČNEJ PRÁCE

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Študijný program: aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
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Názov: Aktívna percepcia v robotickom videní
Active perception in robotic vision

Anotácia: Proces percepcie v robotickom kontexte je charakteristický aktívnou interakciou s objektmi počas učenia. Tieto procesy súčasne umožňujú "lacné" generovanie rozmanitých tréningových dát, ktoré sa dajú využiť na rôzne úlohy, napríklad robustne rozpoznávať objekty.

Cieľ: 1. Štúdium literatúry v oblasti aktívnej percepcie.
2. Návrh modelu neurónovej siete ktorá umožní robotovi skúmať objekt držaný v ruke v rôznych uhloch.
3. Vyhodnotenie funkčnosti a úspešnosti modelu a návrh možných vylepšení.

Literatúra: End to end active perception, I. Kostrikov, D. Erhan & S. Levine, NIPS 2016, http://www.dumitru.ca/files/DL_symposium_active_perception.pdf
Active Perception and Representation for Robotic Manipulation, Y.Zaky, G. Paruthi, B. Tripp, J. Bergstra, <https://arxiv.org/abs/2003.06734>

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prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2023

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Bc. Patrik Modrovský

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do later

Abstract

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Obsah

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Motivation

nic moc zatiaľ

Kapitola 1

Introduction

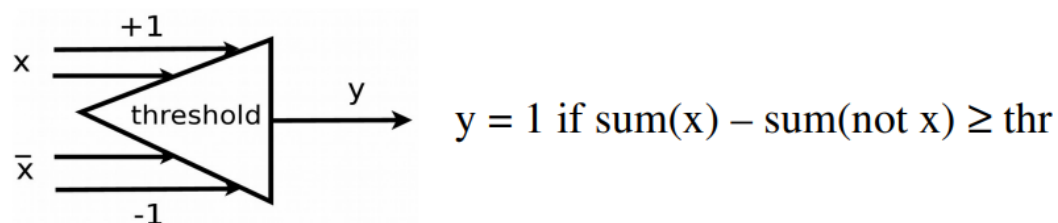
1.1 Neural networks

An Artificial Neural Network (ANN) is a processing system made from large number of interconnected simple processing elements - neurons. Architecture ANNs was inspired by structure of biological brain - both brain and ANN process information similarly and learns from examples.

This creates different approach to problem solving compared to conventional computing. Instead of predictable algorithmic approach, ANNs allows solving problems without understanding of problem, but with lower accuracy. [4]

1.1.1 History

In 1943 neurophysiologist Warren McCulloch and mathematician, Walter Pitts developed first model of ANN. Neurons in their models had fixed threshold and were able to simulate simple logic functions. [4]



Obr. 1.1: Model of an artificial neuron according to McCulloch and Pitts [2]

After initial period of enthusiasm and few other contributions to the field, period of frustration followed. During that time there were only few researchers continued work on solving problems with neural networks. During this time Klopff developed basis for learning and Werbos developed back-propagation algorithm.

This period lasted up to the 1980s, when multiple new contributions renewed wider interest. Hopfield described recurrent neural networks and presented paper *Neural Networks and Physical Systems with Emergent Collective Computational Abilities*. [4]

Neural networks become recently much more publicly known, caused by advances in computing power allowing much bigger number of neurons. It allowed for much more complex tasks as image generation from natural language prompts or chatbots capable of deeper conversations.

1.1.2 Different models of neural networks

There are hundreds of different models and architectures of neural networks with different advantages and disadvantages for different purposes. Some of the most used architectures are:

- Single perceptrons
- Multi-layer perceptrons
- Convolutional neural network
- Recurrent neural network
- Self-Organizing Maps
- Radial Basis Function Network
- Hopfield autoassociative memory

In practise, multiple architectures are often combined to achieve desired outcome. [2]

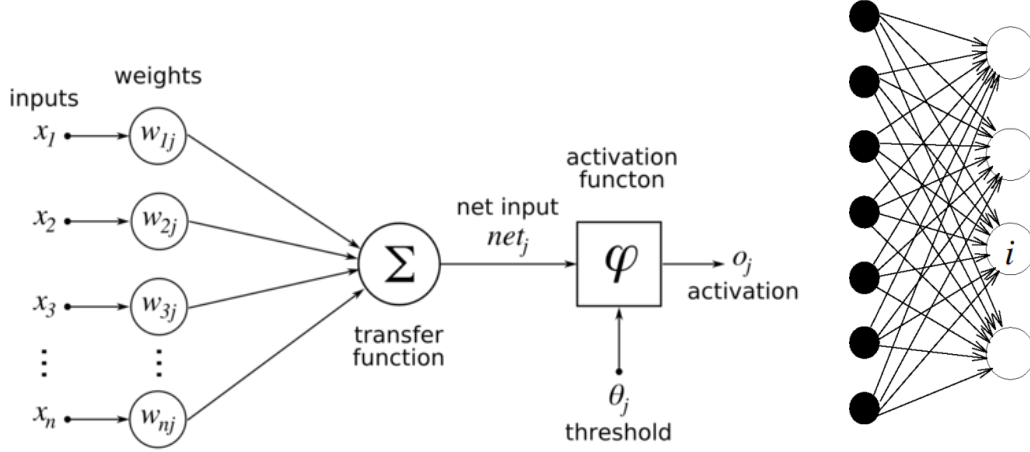
Simple perceptron

It was proposed in 1958 by American psychologist, F. Rosenblatt. It is more general model, with free parameters, stochastic connectivity and threshold elements. [2]

Activation of this perceptron consists of weighted sum of inputs, from which we subtract threshold θ (1.1). Result is then run through activation function, either uni/bipolar for discrete perceptron or sigmoid for continuous perceptron. In practise, threshold is expressed as another weighted input x_{n+1} , with static value of -1, commonly called bias (1.2).

$$o_j = f\left(\sum_{i=1}^n w_{ij}x_i - \theta_j\right) \quad (1.1)$$

$$o_j = f\left(\sum_{i=1}^{n+1} w_{ij}x_i\right) \quad (1.2)$$



Obr. 1.2: Model of simple perceptron and layer of simple perceptrons [2]

Learning rule for adjust weights in discrete perceptron:

$$w_j(t+1) = w_j(t) + \alpha(d - y)x_j \quad (1.3)$$

And for continuous perceptrons:

$$w_j(t+1) = w_j(t) + \alpha(d^{(p)} - y^{(p)})f'x_j \quad (1.4)$$

Where α (*Alpha*) represents learning rate.

Simple perceptron can solve only lineary separable classes. This caused loss of interest in neural networks in many researchers 1970s as many real problems are lineary non-separable, thus simple perceptron cannot be used to solve them.

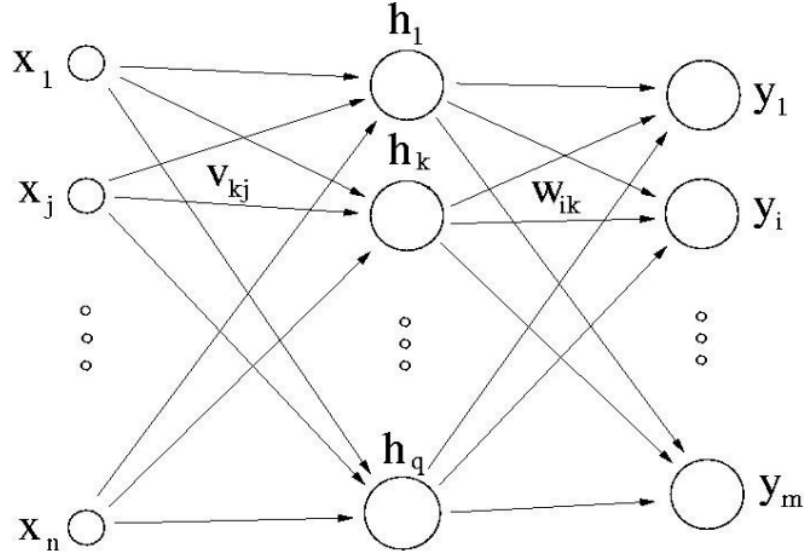
Multi-layer perceptron

Multi-layer perceptron(MLP) is probably one of most used architectures today. It is generalisation of simple perceptrons and it contains additional hidden layers. MLP originates from *Rumelhart & McClelland: Parallel distributed processing* but was earlier described by Werbos in 1974. It is response to critique on perceptrons.[2]

In MLP Activations are similar to activations in simple perceptron. Their distinction comes from addition of hidden layers and additional weights. Due this change, output layer is calculated from hidden layer, instead of inputs.

$$h_k = f\left(\sum_{j=1}^{n+1} v_{kj}x_j\right) \quad (1.5)$$

$$y_i = f\left(\sum_{k=1}^{q+1} w_{ik}h_k\right) \quad (1.6)$$



Obr. 1.3: Model of multi-layer neural network [2]

Activation functions on hidden layers doesn't have similar constraints as output function, but linear function should not be used, as it doesn't add any depth to network. Common activation functions are sigmoid or hyperbolic tangent function.

Learning of MLP uses back-propagation algorithm with different equations for hidden and output layers. Weight adjustments are calculated backwards from output layers to the first hidden layer.[2]

Equation for calculating Hidden-output weights:

$$w_{ik}(t+1) = w_{ik}(t) + \alpha \delta_i h_k \text{ where } \delta_i = (d_i - y_i) f'_i \quad (1.7)$$

And for Input-hidden weights:

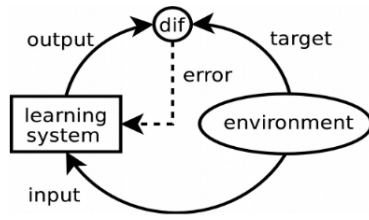
$$v_{kj}(t+1) = v_{kj}(t) + \alpha \delta_k x_j \text{ where } \delta_k = \left(\sum_i w_{ik} \delta_i \right) f'_k \quad (1.8)$$

1.2 Reinforcement learning

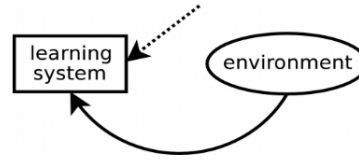
In the realm of artificial intelligence, learning methods can be categorized into three types: supervised learning, unsupervised learning, and reinforcement learning, each having its own uses and advantages.

Supervised learning learns from a labelled dataset, where each input has a corresponding desired output. The goal is generalize and make accurate predictions for new, unseen data. This method is commonly used for regression and classification tasks, where the algorithm iteratively adjusts its parameters to minimize the difference between predicted and true values. Accuracy of predictions is heavily dependent of quality

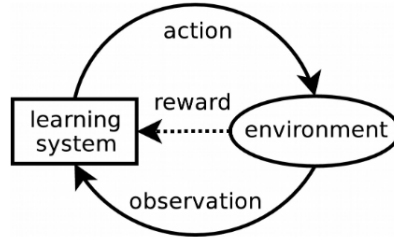
supervised (with teacher)



unsupervised (self-organized)



reinforcement learning (partial feedback)



Obr. 1.4: Different categories of AI learning methods [2]

of dataset.

Unsupervised learning learns on unlabelled datasets. It discovers inherent patterns or structures within the data. Common tasks include clustering similar data points or reducing dataset dimensionality. Unsupervised learning is used for exploring large datasets, uncovering hidden relationships, and helping in data visualization.

Reinforcement learning learns from interaction with environment through actions and rewards. Its goal is mapping environment states to actions order to maximize long term reward. It is used in robotics, self-driving cars or playing games. Two main characteristics of RL are trial and error and delayed reward.

1.2.1 Introduction to RL

Reinforcement learning is mapping actions for every state in order to maximize rewards. Rewards are not know to the learner and have to be discovered by trying them. In more complex cases, actions affects all subsequent states and rewards.

Reinforcement learning simultaneously refers to problem, class of solution methods and field that studies this problem.

In RL problem, a learning agent must be able to sense state of the environment, take actions that can affect said state and have goal or goals relating to the state of environment.

Reinforcement learning is different from supervised learning. Supervised learning learns from already mapped set of data, where every label has a correct action and model tries to generalise for previously unseen data, for example correctly categorize

animal in picture or calculate y from input x . Despite its usefulness, it is inadequate for learning from interactions, as it is often impractical to obtain representative examples of desired action for all situations in which the agent has to act. Agent must be able to learn from its own experience in order to find best action for every state.[6]

Literatúra

- [1] Arden Dertat. Applied deep learning - part 4: Convolutional neural networks. 2017.
- [2] Igor Farkaš. Neural networks. page 192. Knížničné a edičné centrum FMFI UK, 2011.
- [3] Dominique A. Heger. An introduction to artificial neural networks (ann)-methods , abstraction , and usage. 2015.
- [4] Bohdan Macukow. Neural networks – state of art, brief history, basic models and architecture. In Khalid Saeed and Władysław Homenda, editors, *Computer Information Systems and Industrial Management*, pages 3–14, Cham, 2016. Springer International Publishing.
- [5] Hai Nguyen and Hung La. Review of deep reinforcement learning for robot manipulation. In *2019 Third IEEE International Conference on Robotic Computing (IRC)*, pages 590–595, 2019.
- [6] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. The MIT Press, second edition, 2018.
- [7] Youssef Zaky, Gaurav Paruthi, Bryan Tripp, and James Bergstra. Active perception and representation for robotic manipulation. *CoRR*, abs/2003.06734, 2020.